Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/09/2006
PI Name:	T'ien, James S. Ph.D.		
Project Title:	Combustion of Solid Fuel in Very Low Speed Oxygen Streams		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jst2@case.edu	**Fax:**	FY 216 368-3007
PI Organization Type:	UNIVERSITY	**Phone:**	4408295283
Organization Name:	Case Western Reserve University		
PI Address 1:	Mechanical and Aerospace Engineering		
PI Address 2:	620 Glennan Hall, 10900 Euclid Avenue		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44106-1712	**Congressional District:**	11
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/1998	**End Date:**	10/31/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Sacksteder, Kurt R	**Contact Phone:**	216-433-2857
Contact Email:	Kurt.R.Sacksteder@grc.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 10/31/2009 (previously 3/31/2008) per PI (Ed., 7/27/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sacksteder, Kurt (NASA Glenn Research Center) Ferkul, Paul (National Center for Microgravity Research)		
Grant/Contract No.:	NNC04AA58A		
Performance Goal No.:	2.4		
Performance Goal Text:			
Task Description:	This task consists of ground-based experiments using low-gravity aircraft and continued development of a comprehensive predictive model of flame spread and extinction over solid fuels. The overall objectives of the research program are to understand the flame spread and extinction phenomena of solids in low-speed forced concurrent flows in microgravity and in upward buoyant flows in partial gravity . Specifically, we are interested in knwing how gravity affects the flammability limits and spread behaviour of solid materials and how to infer the combustion behaviour of these materials in low gravity from test results in normal gravity. The significance of the task is to understand the material flammability in reduced gravity and material screening tests for low gravity applications; this can contribute to a safer space exploration mission.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study on the basic mechanism of material flammability contributes to the fire safety of human both on earth and in space travel and exploration -The PI Has served as a member of the National Academy of Engineering (NRC)committee to identify innovative research needs to foster fire safety in the US.
Task Progress:	1. A preventive approach to fire safety is proper material selection based on the flammability characteristics. One such measure of flammability is the Limiting oxygen Index (LOI). This is a commonly used numerical flammability index for relative grading and selecting materials. The test measures the extinction limit of a downward spreading flame over a finite size (rectangular slab or rod) fuel specimen. In this work [1], a two-dimensional numerical model was used to study the flame stabilization and extinction characteristics over a thick model solid fuel, which resembles a specimen burning near extinction limit in an LOI testing procedure. The computations were preformed for mixed buoyant-forced flow environment prevailing in the LOI test at normal gravity. At sufficiently high oxygen level (far away from flame extinction limit) the flame is anchored on the sides of the fuel slab (side-stabilized flame), but as the oxygen level is reduced, at a certain value the flame stabilization point abruptly shifts from the sides of the specimen to the wake region (wake flame). The structural details of these two modes of flame stabilization are discussed. The computations also show that the shift in the flame anchoring position with oxygen level exhibits hysteresis, i.e. multiple flame solutions may exist for a given environmental condition of flow and oxygen. The flame extinction limits and flame-shift limits (side-stabilized flame to wake flame and vice versa) in ambient oxygen percentage are presented for different forced flow velocities. Over the velocity range studied here, the side-stabilized flame (similar to flames over of thin fuels) becomes less flammable (higher LOI) at higher velocity whereas, the wake stabilized flame becomes more flammable (lower LOI) with increase in forced flow velocity. To understand the implication of this earth based measurement to material selection for space application, additional computations were performed for pure-forced flow environment in zero gravity. Stabilization and extinction behavior of the flames at normal gravity is compared with that in zero gravity. The computations in pure forced flow environment show that the two modes are also present in zero gravity environment but only above certain minimum flow velocity. Below this velocity only single flaming solution was obtained. In the present computations, the LOI values at zero gravity were lower than those at normal gravity. For zero gravity the LOI based on side stabilized flame (for thin fuels) shows a non-monotonic behavior, there is a minimum LOI which for the present set of property and kinetic parameter values occurs at forced velocity of about 3-5cm/s. The LOI based on wake flame extinction, at zero gravity decreases rapidly with flow velocity for small velocities but with further increase in velocity the LOI value stays approximately constant. 2. The Limiting Oxygen Index (LOI) apparatus has traditionally been used to quantify flammability characteristics of the materials numerically. The testing procedure consists determining the minimum concentration of oxygen in a mixture with nitrogen that when passed through the column will support combustion of solid fuel ignited from the top. Although a simple device, the flow pattern inside the device could be complicated by buoyant velocity induced under the influence of earth's gravitational force field. This in turn could affect the measured LOI value. So far the experimental studies are limited to global measurements and there is no detailed study on this device. In this work [2], the LOI test apparatus is numerically studied using a three-dimensional model for downward flame spread over a film type solid fuel. The model capability is exploited to investigate the mixed buoyant-forced flow field in the test column, the solid fuel combustion details and flame extinction behavior. The computations show that buoyancy effects can result in the ambient air from outside to be drawn into the column. The degree of entrainment decreases as the gas mixture mass feed rate at the base of the column is increased. Over the velocity range 0.5cm/s-15cm/s, the LOI values are almost constant to change in the column base inlet velocity. In the absence of buoyant force (which has implication to the zero gravity space environment) the flow field is drastically different from that in normal gravity, and the computed LOI values for the chosen fuel are lower than those for normal gravity cases and exhibited a non monotonic trend with a minimum LOI. The LOI was found to be independent of specimen width for width greater than 1cm. 3. In the enclosed environment of a manned space habitat, fire safety becomes a major concern. Though various combinations of pressure and oxygen can be used to support human habitation, the bulk of flammability testing for materials is done at standard atmospheric conditions. This study focuses on the flammability and flame spread in these potential space habitat environments [3 and Kleinhenz's PhD thesis]. Pressures were kept below 1 atm, Oxgyen/Nitrogen mixtures between 21% and 30%, and gravity levels of 1 g/ge (Earth), 0.38 g/ge (Martian), and 0.16 g/ge (Lunar) were used.Two materials were examined. Part 1 of this report discusses Kimwipes®, a thin tissue paper, which was better suited for the reduced gravity experiments. A scaling relation that predicts upward spread rate as a function of pressure, gravity, and width, is introduced and experimentally verified. Nomex® fabric, practical material commonly used in the space program, is the focus of Part 2. Nomex has a lower oxygen limit for upward spread than for downward spread. A unique phenomenon was observed during experimentation, whereby the continually elongating flames broke off in the middle into two smaller upward propagating flames. One flame would then extinguish, and the remaining flame would repeat this cycle. The process was documented in detail and is believed to be a function of a two stage pyrolysis occurring in the fuel.In order to enhance the capability of flame-spreading model as a predictive tool, especial in microgravity, we have performed radiation property measurement, spreading tests in normal gravity, and model calibration exercises for a specific cotton-fiberglass composite fabric. The paper gives details of how these are done and some of the special features of the burning of a composite fuel [3]. 4. Using a detailed three-dimensional model, the effects of gas-phase chemical kinetic rate and pressure on flame-spread rates are investigated for both buoyant upward and concurrent forced flow flame spread over thin solids [4]. For wide (two-dimensional) samples, varying the chemical kinetic rate alone has a negligible effect on spread rate in forced concurrent flow spread; it has a small but finite influence on upward buoyant spread rate. The latter is thought to be through the increasing buoyant velocity at higher flame temperature when the kinetics becomes faster. In narrow samples, the influence of kinetics is increased, thought due to the side quenching influence from kinetic rates. However, pressure has a much larger influence on concurrent flame spread rate than that from kinetics alone. This is primarily through the pressure effect on flame standoff distance that affects the fuel burning and heat release rate. Based on this sensitivity study, the possibility of pressure modeling is examined. Comparing with a previously proposed scheme of pressure modeling derived from reduced gravity experiment, a small discrepancy has been found. It is suggested that this discrepancy is due to the lack of flame radiation in the flame spread model. 5. A three-dimensional concurrent flame spread model that includes a gray gas radiation model has been solved for 5 cm/s purely forced flow over a thin cellulosic fuel [5]. Cases are compared for full Planck mean radiation, no radiation, and an intermediate case which is thought to better represent the radiative energy interaction. Results show that the radiation decreases the flame temperature, flame length and spread rate. Flame profiles, heat fluxes, and additional

structural details are presented.

6. The importance of radiation effects in the combustion of solid fuels means that surface radiative properties must be well known under a variety of conditions to allow accurate numerical modeling. This paper [6] and Pettegrew's PhD thesis provide partial results (both spectral and band-integrated) from an experimental effort that determined the radiative properties of a thin cellulosic fuel as a function of viewing angle at discrete levels of pyrolysis.

7. Flame spread over a thermally-thin solid fuel is numerically investigated under various boundary conditions using a detailed two-dimensional, opposed-flow model with gas-phase radiation and the latest Kimwipe pyrolysis kinetics data [7]. The effects of solid virgin fuel temperature, external radiation and fuel vapor doping in opposed air flow are systematically studied because of their importance in room fires (flashover) and in spacecraft fires (overheated equipments). The numerical results show that all of these elemental factors involved in a room fire can enhance the flame spread rate but their mechanisms of enhancement are not the same. This paper will present the detailed flame responses to each individual factor [7].

8. Additional reduced gravity experiments on flame spreading and upward oxygen flammability limits have recently been carried out on C-9 mission aboard the KC135 airplane in March 2006. The data are still being analyzed. The first part of the data should reinforce the pressure-gravity modeling relationship proposed previously. The second part of the data will be useful in assessing the effect of partial gravity on upward flammability limit and the effect of sample length (relative to width) in performing the upward burning experiments.

1. Kumar, A. and T'ien, J. S.: A Computational Study of Low Oxygen Flammability Limit for Thick Solid Samples, Combustion and Flame, in press.

2. Kumar, A. and T'ien, J. S.: Numerical Modeling of Limiting Oxygen Index Apparatus for Film Type Fuels, to be submitted to Fire Safety Journal.

3. Kleinhenz, J., T'ien. J. and Pettegrew, R.: Flammability and Flame Spread of Nomex Fabrics in Space Habitat Environments, to be presented at the 2006 Technical Meeting of the Central States Section of The Combustion Institute, May 21-23, 2006, Cleveland, Ohio.

4. Hsu, S.-Y. and T'ien, J..: Concurrent-Flow Flame Spread over Solids: Effects of Pressure and Chemical Kinetics, to be presented at the 2006 Technical Meeting of the Central States Section of The Combustion Institute, May 21-23, 2006, Cleveland, Ohio.

5. Feier, I., T'ien, J. and Bedir, H.: Effect of Flame Radiation on Three-dimensional Concurrent Flame Spread Over Solids, to be presented at the 2006 Technical Meeting of the Central States Section of The Combustion Institute, May 21-23, 2006, Cleveland, Ohio.

6. Pettegrew, R., T'ien, J. and Street, K.: Spectral Radiative Characteristics of a Thin Cellulosic Fuel at Discrete Levels of Pyrolysis, to be presented at the 2006 Technical Meeting of the Central States Section of The Combustion Institute, May 21-23, 2006, Cleveland, Ohio.

7. Han, B., T'ien, J. and Kumar, A. Effects of Virgin Fuel Temperature, External Radiation and Fuel Vapor Doping in Opposed Air on Flame Spread, to be presented at the 2006 Technical Meeting of the Central States Section of The Combustion Institute, May 21-23, 2006, Cleveland, Ohio.

Bibliography Type:	Description: (Last Updated: 09/19/2025)
Articles in Peer-reviewed Journals	Kumar A, T'ien JS. "A computational study of low oxygen flammability limit for thick solid slabs." Combustion and Flame 2006 Jul;146(1-2):366-78. http://dx.doi.org/10.1016/j.combustflame.2006.02.008 , Jul-2006
Dissertations and Theses	Pettegrew RD. "Radiative Characteristics of a Thin Cellulosic Fuel at Discrete Levels of Pyrolysis: Angular, Spectral, and Thermal Dependencies." Ph.D. Dissertation, Case Western Reserve University, January 2006. , Jan-2006
Dissertations and Theses	Kleinhenz JE. "Flammability and Flame Spread of Nomex® and Cellulose in Space Habitat Environments. " Ph.D. Dissertation, Case Western Reserve University, May 2006. , May-2006
Papers from Meeting Proceedings	Kleinhenz J, T'ien J, Pettegrew R. "Flammability and Flame Spread of Nomex Fabrics in Space Habitat Environments." Spring Technical Meeting of the Central States Section of The Combustion Institute, Cleveland, Ohio, May 2006. Spring Technical Meeting of the Central States Section of The Combustion Institute, Cleveland, Ohio, May 2006. , May-2006
Papers from Meeting Proceedings	Hsu S-Y, T'ien J. "Concurrent-Flow Flame Spread over Solids: Effects of Pressure and Chemical Kinetics." Spring Technical Meeting of the Central States Section of The Combustion Institute, Cleveland, Ohio, May 2006. Spring Technical Meeting of the Central States Section of The Combustion Institute, Cleveland, Ohio, May 2006. , May-2006
Papers from Meeting Proceedings	Feier I, T'ien J, Bedir H. "Effect of Flame Radiation on Three-dimensional Concurrent Flame Spread Over Solids." Spring Technical Meeting of the Central States Section of The Combustion Institute, Cleveland, Ohio, May 2006. Spring Technical Meeting of the Central States Section of The Combustion Institute, Cleveland, Ohio, May 2006. , May-2006
Papers from Meeting Proceedings	Pettegrew R, T'ien J, Street K. "Spectral Radiative Characteristics of a Thin Cellulosic Fuel at Discrete Levels of Pyrolysis." Spring Technical Meeting of the Central States Section of The Combustion Institute, Cleveland, Ohio, May 2006. Spring Technical Meeting of the Central States Section of The Combustion Institute, Cleveland, Ohio, May 2006. , May-2006

Papers from Meeting Proceedings	Han B, T'ien J, Kumar A. "Effects of Virgin Fuel Temperature, External Radiation and Fuel Vapor Doping in Opposed Air on Flame Spread." Spring Technical Meeting of the Central States Section of The Combustion Institute, Cleveland, Ohio, May 2006. Spring Technical Meeting of the Central States Section of The Combustion Institute, Cleveland, Ohio, May 2006. , May-2006